

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2025-26 (NEP)

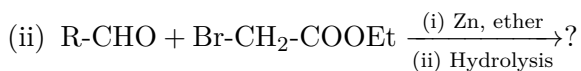
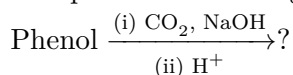
CHEMISTRY**Paper Code: CHEMJ32 : Organic Chemistry-I**

Ref. No.: CECONF/N/2025-26/056

Time: 2½ Hours

Maximum Marks: 70

Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.

Question 1.(a) Complete the following reactions with mechanisms: [2 × 2½ = 5](b) Complete the following reaction with mechanism: [2½](c) Classify the following molecules (I-IV) as aromatic, anti-aromatic or non-aromatic: [2]

(i) Pyridine / Pyrrole derivative

(ii) Boron heterocyclic 5-membered ring

(iii) Boron-nitrogen 7-membered ring

(iv) Tropone

(d) Comment on the aromaticity and dipole moment (μ) of the given below compounds I and II: [2½]

(i) Naphthalene / Azulene derivative

(ii) Fulvalene derivative

(e) Complete the following reaction with mechanism: [2]**Question 2.**(a) Among the following compounds (I and II) which one have the lowest free energy barrier for cis-trans isomerisation and why? [3](b) Draw the Molecular Orbital picture of benzene. [3](c) Discuss the mechanism of nitration of benzene with energy profile diagram. Provide evidence in support of its mechanism. [6](d) Comment on the nitration of Anisole ($\text{C}_6\text{H}_5\text{-OCH}_3$) using either mixed acid condition ($\text{HNO}_3 + \text{H}_2\text{SO}_4$) or N_2O_5 (fuming). [2]

Question 3.

- (a) Discuss the Sulfonation of benzene with mechanism. [3]
- (b) Discuss the mechanism of Friedel-Crafts Alkylation. Also, point out its limitations. [3+3]
- (c) Compare the relative reactivity of benzene, nitrobenzene, toluene and phenol towards electrophilic aromatic substitution reaction. [2]
- (d) What would be the orientation (major product formed) when nitration of the following compounds I-VI were performed separately: [3]
- (i) Benzoic acid (Ph-COOH)
 - (ii) Toluene (Ph-CH₃)
 - (iii) Nitrobenzene (Ph-NO₂)
 - (iv) Biphenyl
 - (v) Benzophenone (Ph-CO-Ph)
 - (vi) Phenol (Ph-OH)

Question 4.

- (a) Provide the mechanism of Diazotization of aniline using HCl + NaNO₂. [4]
- (b) How will you prepare m-dibromobenzene starting from benzene? [2]
- (c) How will you prepare m-chlorobenzaldehyde starting from benzaldehyde? [2]
- (d) How will you prepare 1,3,5-tribromobenzene starting from aniline? [2]
- (e) What will be the product when nitrobenzene undergo reduction under neutral condition? [2]
- (f) What will be the major products in the following reaction? [2]
- $$\text{ArN}_2^+\text{Cl}^- + \text{Phenol} \xrightarrow{\text{OH}^-} (\text{A})? \quad ; \quad \text{ArN}_2^+\text{Cl}^- + \text{Aniline} \xrightarrow{\text{H}^+} (\text{B})?$$

Question 5.

- (a) Write the synthesis and application of DDT. [3]
- (b) What is BHC and how can you synthesize it? [3]
- (c) Explain benzyne mechanism with suitable example. Provide evidence in support of its mechanism. [5]
- (d) Complete the following reaction with suitable mechanism: [3]
- $$2\text{-(chloromethyl)pyridine} + \text{PhLi} \rightarrow ?$$

Question 6.

- (a) Discuss the acid-catalysed Mannich reaction with mechanism. [4]
- (b) Discuss the given below Baylis-Hillman reaction with mechanism: [4]
- $$\text{Ph-CHO} + \text{CH}_2 = \text{CH-COOEt} \xrightarrow{\text{DABCO}} ?$$
- (c) Explain the relative reactivity of acid chloride, acid anhydride, amide and ester towards nucleophilic addition with suitable example. [4]

- (d) Complete the following reaction with mechanism: [2]
 $\text{Ph-COCH}_2\text{CHO} \xrightarrow{\text{NaOH}} ?$

Question 7.

- (a) Among the following compounds (I and II) which one underwent fast solvolysis and why? [2]
- (b) Comment on the reactivity of given below compounds (I-III) with acetaldehyde: [2]
- (i) $\text{Ph}_3\text{P}^+ - \text{CH}_2^-$
- (ii) $\text{Ph}_3\text{P}^+ - \text{CH}^- - \text{COR}$
- (iii) $\text{Ph}_3\text{P}^+ - \text{Cyclopentadienyl}^-$
- (c) Discuss the Wittig reaction with mechanism. [3]
- (d) Complete the following reaction with suitable mechanism: [2]
1,5-dicarbonyl compound $\xrightarrow[\text{(ii) H}_2\text{O}]{\text{(i) OH}^-}$?
- (e) Provide the synthetic sequence for the following conversion: [3]
 $\text{H}_3\text{C-CHO} \xrightarrow{?} \text{Pentaerythritol } [\text{C}(\text{CH}_2\text{OH})_4]$

Question 8. Discuss the mechanism of the following reactions: [4 × 3½ = 14]

- (a) Hell-Volhard-Zelinsky reaction
- (b) Perkin reaction
- (c) Haloform reaction
- (d) Reimer-Tiemann reaction

Complete the reaction: $\text{Ph-COOH} \xrightarrow{\text{SOCl}_2} ? \xrightarrow{\text{CH}_2\text{N}_2} ?$ [2]
